

Software Engineering Lab 1

Requirements Engineering & UML Use-Case Modelling

Problem Statement #46

Remote Team Time-Tracking & Project Approver

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Use Case: UC-02 — Submit Timesheet

Primary Actor: Remote Developer

Supporting Actor: Engineering Manager

1. Preconditions

1. The Remote Developer is authenticated in the system.
2. The developer has logged time against project tasks.
3. The timesheet contains valid project/task information.
4. The timesheet does not exceed the permitted 24 hours for a single calendar day.

2. Postconditions

1. The timesheet is successfully submitted.
2. The timesheet status changes to **Pending Approval**.
3. The Engineering Manager can view the submitted timesheet.
4. The submitted hours are available for project tracking and budget calculations.

3. Main Success Scenario

1. Remote Developer logs into the time-tracking system.
2. Developer reviews the recorded hours for the relevant project tasks.
3. Developer selects **Submit Timesheet**.
4. System validates the timesheet details.
5. System checks that the submitted hours do not exceed **24 hours in a single calendar day**.
6. System records the timesheet submission.
7. System changes the timesheet status to **Pending Approval**.
8. System makes the submitted timesheet available to the Engineering Manager.
9. Engineering Manager receives/Views the pending timesheet.
10. Use case ends successfully.

4. Alternate Flow — Daily Limit Exceeded

5a. Submitted hours exceed 24 hours in a calendar day.

1. System detects that the submitted hours exceed the 24-hour limit.
2. System prevents the timesheet from being submitted.
3. System displays an error message indicating that the daily limit has been exceeded.
4. Remote Developer corrects the recorded hours.
5. Developer resubmits the timesheet.
6. System repeats the validation process.